Interpretable Credit-Risk Classification Using Rule Induction and Noise Robustness

Naga Sampath Maddineni
M.S., Artificial Intelligence - Machine Learning
Department of CSE
University of North Texas
Denton, Texas, USA
11751165

Abstract - Credit-risk classification is a consequential binary decision because the same prediction affects lender exposure and an applicant's access to credit. This study compares a pruned decision tree, PART, and RIPPER/JRip on the Statlog German Credit dataset using a common preprocessing pipeline, repeated cross-validation, statistical testing, and a controlled label-noise experiment. JRip obtains the highest clean-data mean accuracy (0.7238) and noisy-data mean accuracy (0.6881), while the decision tree has the smallest clean-data variance. The results show that accuracy, rule compactness, stability, and label-noise robustness provide complementary evidence for selecting an interpretable classifier.

Index Terms - credit risk, interpretable classification, decision tree, PART, RIPPER, JRip, t-SNE, label noise, repeated cross-validation.

I. INTRODUCTION

A. Background and Motivation:

Credit-risk assessment separates applicants who are likely to repay from those who may default. A classification error can increase a lender's expected loss or deny an applicant access to useful credit. A practical model must therefore support review: analysts need to understand which applicant characteristics drove a predicted class. Transparent paths and ordered conditions make it possible to inspect an individual prediction and question an unexpected rule. The German credit data are a compact benchmark with mixed categorical and numeric attributes, including checking status, credit history, duration, savings, employment, housing, and loan purpose. The class overlap makes the data suitable for comparing interpretable classifiers with different rule structures.

B. Problem Statement:

Credit-risk classification must distinguish good and bad applicants while remaining understandable to the people who review lending decisions. This study compares three interpretable classifiers and measures how their performance changes when class labels contain noise.

C. Study Objective:

The objective is to compare a pruned decision tree, PART, and RIPPER/JRip under one reproducible experimental design. The comparison records clean-data accuracy, fold-to-fold variance, rule count, statistical differences between model means, and the change in accuracy after 10% of the class labels are flipped. The results identify JRip as the best overall model under the composite ranking, followed by the decision tree and PART.

II. STUDY CONTEXT

A. Credit-Risk Classification:

The input is a version of the Statlog German Credit Data distributed through the UCI Machine Learning Repository [1], [6]. It contains a binary good/bad target and 20 mixed financial and demographic predictors. The target contains 700 good and 300 bad observations. Accuracy is useful for an initial comparison, but the imbalance makes it important to report the complete validation vectors and variation rather than relying on a single aggregate score.

B. Interpretable Rule Learning:

Decision trees represent a prediction as a path from the root to a terminal class. PART converts partial tree structures into an ordered decision list, while JRip implements a RIPPER-style learner that grows and optimizes rules and retains a default class. These representations expose conditions directly and support comparison of compactness, local detail, and stability [2], [3], [5], [8], [9].

C. Exploratory Representation:

t-SNE is used only as an exploratory representation of one-hot encoded and scaled attributes. It helps inspect local neighborhoods and class overlap, but the projected coordinates are not predictive inputs and the axes do not have direct financial meaning [4].

D. Comparative Evaluation:

The study reports mean accuracy, fold-to-fold variance, rule count, and the change in mean accuracy after label noise. ANOVA tests the overall model effect, Tukey's HSD identifies pairwise differences, and the final ranking combines rule compactness, clean accuracy, stability, and robustness.

III. DATA SET PROCESSING AND EXPLORATORY ANALYSIS

A. Dataset Description:

The input file contains 1,000 observations, 20 predictors, and a binary target. Before fitting a model, the script checks file existence, row count, exact 21-column schema, target presence, required columns, and missing values. No missing values are present. These checks prevent an accidental input change from being interpreted as a modeling result.

Property	Description
Observations	1,000
Predictors	20 mixed categorical and numeric attributes
Target	Good or bad credit classification
Class counts	700 good; 300 bad
Initial split	Stratified 70/30 train/test split
Selected features	15 highest chi-squared attributes
Missing values	None detected

B. Preprocessing and Feature Selection:

Duplicate rows are removed before the initial split. Character and logical variables are converted to factors, and low-cardinality integer-coded categorical variables are treated as factors when their codes represent levels rather than measurements. Test-set factor levels are aligned with training levels. Chi-squared association retains the 15 strongest attributes for modeling. One-hot encoding and scaling are used only for t-SNE, while the classifiers retain interpretable categorical conditions.

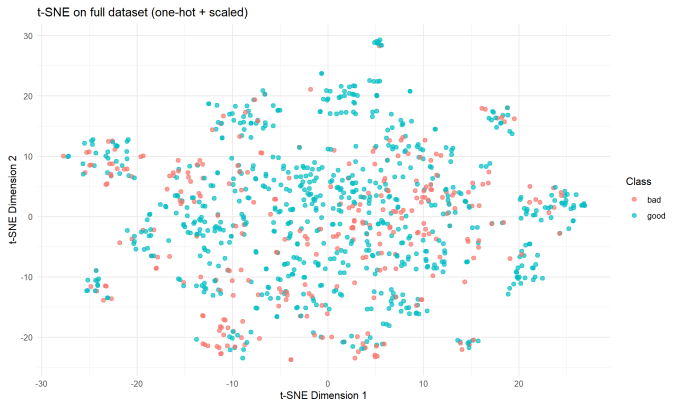


Fig. 1. t-SNE projection of the one-hot encoded and scaled dataset.

The t-SNE plot shows substantial overlap between good and bad cases. Some local neighborhoods show partial concentration by class, but there is no clean global separation. This indicates that useful predictions depend on combinations of conditions rather than a simple visual boundary. The projection is descriptive and should not be used to infer that one classifier is superior.

C. Pruned-Tree Interpretation:

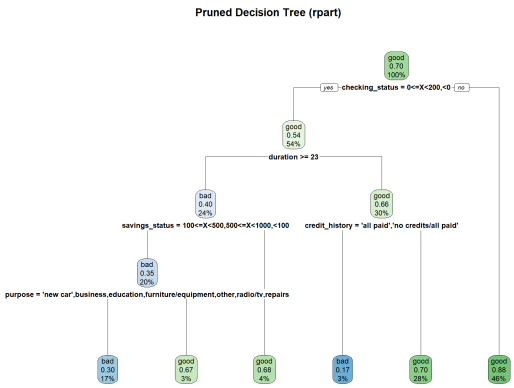


Fig. 2. Pruned decision tree generated by the current workflow.

The pruned tree is organized primarily around checking status and duration, with savings status, purpose, and credit history appearing on lower branches. Each terminal path gives a class prediction and local support. Pruning removes branches that do not justify their complexity under the one-standard-error rule, leaving six paths that remain visually traceable.

IV. MACHINE-LEARNING METHODOLOGY

A. Problem Formulation:

Let x_i denote the selected attributes for applicant i and y_i belong to {good, bad}. Each learner estimates a function that assigns one of the two classes. Accuracy is the proportion of correctly classified held-out cases in each validation fold. All three methods receive the same selected modeling frame and resampling indices, so their differences reflect the learners and their induced decision structures.

B. Models Considered:

The decision tree uses information-based splitting, a minimum split size of 10, and an initial complexity parameter of 0.001 before pruning by the one-standard-error rule. PART induces an ordered partial decision list. JRip grows and optimizes an ordered RIPPER rule set, removes unnecessary conditions, and ends with a default class. The tree is hierarchical, PART is a longer ordered list, and JRip is a compact ordered rule base.

Classifier	Rule count	Representative form
Decision Tree	6	Tree path to terminal class
PART	44	Ordered conjunctive decision list
RIPPER/JRip	4	Compact ordered rule set

C. Training and Reproducibility:

The initial split uses seed 123. Clean cross-validation uses 10 folds and three repeats with seed 3333, producing 30 accuracy observations per classifier. A noisy dataset is created by flipping 100 labels without replacement using seed 2026. The same preprocessing, resampling design, and model settings are then applied. Fixed seeds make the fold vectors and exported rules reproducible under the installed R/RWeka environment.

D. Model Outputs:

The decision-tree file contains six terminal-class rules, the PART file contains 44 ordered rules, and the JRip file contains four rules including its default behavior. Retaining these rule files permits a direct audit of compactness and makes the conditions readable without rerunning the software.

V. VALIDATION AND STATISTICAL ANALYSIS

A. Cross-Validation Design:

Each model produces 30 fold-level accuracy values for clean data and 30 after label noise. The mean is the sum of fold accuracies divided by n, and sample variance divides the squared deviations by n minus 1, with n=30. Lower variance indicates greater stability, but repeated folds remain correlated because observations can occur in several training partitions.

B. Clean-Data Accuracy Vectors:

The complete vectors make the summary statistics auditable. The decision tree values are concentrated near 0.70, PART has a wider spread, and JRip contains several higher values while still varying across folds.

Fold	DT	PART	JRip
1	0.7143	0.6571	0.7000
2	0.6714	0.6143	0.8000
3	0.7143	0.7000	0.7857
4	0.7143	0.6714	0.7286
5	0.7000	0.7429	0.7286
6	0.7000	0.7857	0.6857
7	0.7000	0.6571	0.6857
8	0.7000	0.8000	0.6286
9	0.7000	0.6714	0.7571
10	0.6571	0.6571	0.7143
11	0.6714	0.6857	0.7143
12	0.7000	0.7571	0.7000
13	0.7000	0.6714	0.7714
14	0.7000	0.6571	0.7571
15	0.7000	0.6571	0.7571
16	0.7000	0.6429	0.7286
17	0.7000	0.7000	0.7143
18	0.7000	0.7429	0.6571
19	0.7000	0.7000	0.7429
20	0.7000	0.7429	0.7000
21	0.7000	0.7000	0.6857
22	0.7000	0.7143	0.7143
23	0.7286	0.6857	0.7286
24	0.7000	0.7286	0.7000
25	0.7000	0.6857	0.7000
26	0.7000	0.6857	0.7429
27	0.7286	0.6429	0.7286
28	0.7143	0.7429	0.7429
29	0.7000	0.7286	0.7286
30	0.7143	0.6714	0.7857

Table I. Complete clean-data accuracy vectors.

C. Clean Summary and Statistical Tests:

The clean means are 0.700952381 for the decision tree, 0.696666667 for PART, and 0.723809524 for JRip. The corresponding variances are 0.000210181, 0.001951912, and 0.001426226. ANOVA is significant, $F(2,87)=5.34046$, $p=0.0064918$.

Model	Mean	Variance
Decision Tree	0.700952381	0.000210181
PART	0.696666667	0.001951912
JRip	0.723809524	0.001426226

Source	df	SS	MS	F	p
Model	2	.0127755	.00638776	5.34046	.0064918
Residual	87	.1040612	.00119611		

Table II. Clean-data means, variances, and ANOVA.

Comparison	Difference	Lower	Upper	Adjusted p
JRip - DT	.0228571	.0015643	.0441499	.0323891
PART - DT	-.0042857	-.0255785	.0170071	.8809583
PART - JRip	-.0271429	-.0484357	-.0058501	.0087008

Table III. Clean-data Tukey comparisons.

D. Noisy-Data Procedure and Accuracy Vectors:

One hundred target values are selected without replacement and their labels are flipped. Feature values remain unchanged, so the experiment isolates label corruption rather than covariate shift. The models are refit with the same repeated-validation design.

Fold	DT	PART	JRip
1	0.6620	0.6571	0.6571
2	0.6620	0.6286	0.7536
3	0.6714	0.6286	0.7183
4	0.6857	0.6377	0.6714
5	0.6714	0.6197	0.6857
6	0.6667	0.7143	0.6197
7	0.6761	0.5857	0.6000
8	0.5714	0.5714	0.7143
9	0.6714	0.6857	0.6714
10	0.6143	0.6286	0.6286
11	0.6571	0.5857	0.6857
12	0.6620	0.6571	0.7143
13	0.6714	0.6000	0.7286
14	0.6714	0.6000	0.7857
15	0.6571	0.6571	0.6667
16	0.6714	0.6714	0.7143
17	0.6714	0.6714	0.6901
18	0.6429	0.6901	0.7286
19	0.6667	0.6429	0.6479
20	0.6714	0.6714	0.6338
21	0.6714	0.6000	0.6479
22	0.6377	0.6761	0.7714
23	0.6571	0.7286	0.7606
24	0.6143	0.7042	0.6000
25	0.6620	0.5857	0.7143
26	0.6429	0.6232	0.7143
27	0.6620	0.6286	0.6667
28	0.6714	0.7324	0.7286
29	0.6714	0.6901	0.6377
30	0.6714	0.6232	0.6857

Table IV. Complete noisy-data accuracy vectors after 10% label flips.

E. Noise Summary and Statistical Tests:

Under label noise, the decision tree mean is 0.658564333, PART is 0.646557801, and JRip is 0.688098251. The mean drops are 0.042388048, 0.050108866, and 0.035711273, respectively. JRip retains the highest mean and loses the smallest amount of accuracy. ANOVA is significant, F(2,87)=8.35783, p=0.00047847.

Model	Noise mean	Noise variance	Drop
Decision Tree	0.658564333	0.000540267	0.042388048
PART	0.646557801	0.001960454	0.050108866
JRip	0.688098251	0.002420443	0.035711273

Source	df	SS	MS	F	p
Model	2	.0274202	.01371009	8.35783	.00047847
Residual	87	.1427137	.00164039		

Table V. Noisy-data means, variances, drops, and ANOVA.

Comparison	Difference	Lower	Upper	Adjusted p
JRip - DT	.0295339	.0045982	.0544696	.0160426
PART - DT	-.0120065	-.0369422	.0129292	.4873439
PART - JRip	-.0415405	-.0664761	-.0166048	.0004265

Table VI. Noisy-data Tukey comparisons.

F. Validation Interpretation:

The vectors show fold-level behavior, variance summarizes sensitivity to resampling, and ANOVA/Tukey tests quantify whether the model means differ within this design. The label-noise experiment adds a robustness view, but real operational errors may be systematic or class-dependent. JRip's smaller mean drop indicates the strongest average retention, while the decision tree's smaller clean variance indicates the most stable clean estimates.

G. Performance Findings:

The validation evidence gives a consistent ordering for mean accuracy: JRip is first, the decision tree is second, and PART is third. It also separates stability from accuracy: the decision tree has the lowest clean variance even though JRip has the highest mean. Under noisy labels, JRip remains first and PART loses the most accuracy. These findings motivate a composite ranking rather than choosing a model from accuracy alone.

VI. COMPARATIVE RESULTS AND DISCUSSION

A. Accuracy and Robustness:

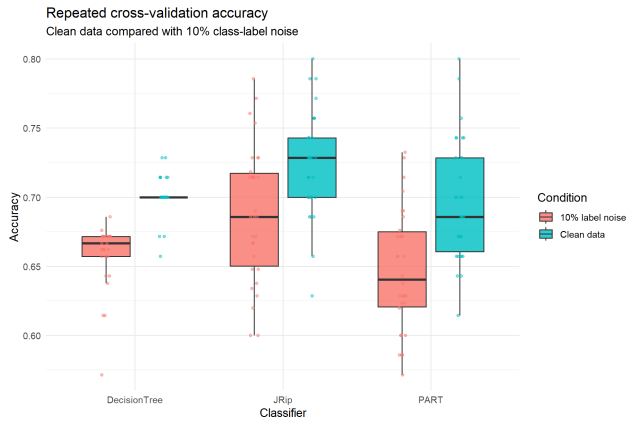


Fig. 3. Current-code repeated cross-validation accuracy under clean data and 10% class-label noise.

JRip has the highest clean-data mean accuracy (0.7238) and noisy-data mean accuracy (0.6881). Its accuracy drop is the smallest, 0.0357. The decision tree has the lowest clean-data variance (0.0002102), making it the most stable classifier across clean validation folds. PART has the lowest mean accuracy in both conditions and the largest drop under label noise. The boxplot shows why a single train/test value would be insufficient: the distributions overlap, but their centers and spreads differ.

B. Rule Perspicuity and Final Ranking:

Rule perspicuity is assessed through rule count and representation. JRip has four rules and a default class, the pruned decision tree exposes six terminal paths, and PART contains 44 ordered rules. Rank 1 is assigned to the best value for each criterion: compactness uses the smallest rule count, accuracy uses the clean mean, stability uses the smallest clean variance, and robustness uses the smallest accuracy drop.

Model	Rules	Accuracy	Stability	Robustness	Mean rank
JRip	1	1	2	1	1.25
Decision Tree	2	2	1	2	1.75
PART	3	3	3	3	3.00

Table VII. Final ranking across rule compactness, accuracy, stability, and robustness.

JRip is the strongest overall choice under this scoring rule because it combines the best accuracy, the smallest noise loss, and the shortest rule base. The decision tree remains attractive when visual paths and stable clean estimates are prioritized. PART is interpretable, but its longer rule list and lower accuracy place it third. These conclusions are specific to the supplied data, preprocessing, seeds, and scoring rule.

C. Limitations:

The experiment uses one benchmark dataset, one target definition, and one controlled label-noise mechanism. Accuracy does not distinguish the costs of false approvals from false rejections, and repeated folds are not a substitute for temporal or external validation. t-SNE is qualitative and can change with preprocessing and initialization. Rule count is also an imperfect proxy for human usability because condition length and domain meaning matter.

VII. CONCLUSION

This study evaluated three interpretable classifiers for binary credit-risk prediction using a common factor-aware preprocessing pipeline, chi-squared feature selection, repeated cross-validation, and controlled label noise. JRip achieved the highest mean accuracy in both conditions and produced the smallest rule base. The decision tree was the most stable on clean validation folds, while PART ranked last on mean accuracy and noise robustness. Future work should evaluate cost-sensitive loss, calibration, subgroup performance, and external validation before applying these models to operational lending decisions.

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